HOUSEHOLD CARBON FOOTPRINT & TARGET TRACKER

Problem Statement 61 – Sustainability & Green Tech

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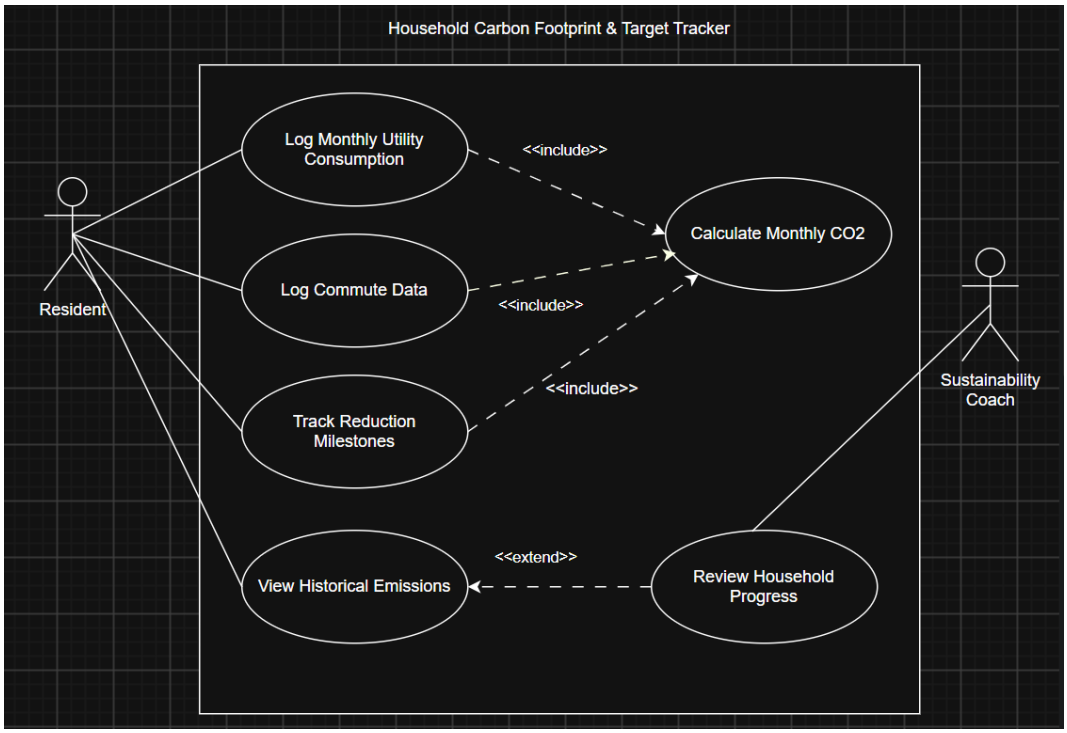
1. Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest household monthly electricity (kWh) and transport (km) logs to compute total monthly carbon equivalent (CO_2e) emissions.	High	Pass: Calculated emissions match EPA conversion factor formulas. Fail: Zero or negative emission values are generated.	To calculate the household's monthly carbon emissions from electricity and transport usage.
FR-002	Functional	The system shall allow residents to log monthly household utility consumption, including electricity and gas usage.	High	Pass: Valid monthly utility values are successfully recorded and stored. Fail: Invalid or incomplete values are accepted.	To collect the household consumption data required for carbon-footprint calculation.
FR-003	Functional	The system shall calculate and display the household's monthly CO_2e emissions based on the recorded activity data.	High	Pass: The system displays a calculated monthly CO_2e value after valid data is submitted. Fail: No result or an incorrect result is displayed.	To provide residents with a measurable value representing their monthly carbon footprint.
FR-004	Functional	The system shall track household progress toward defined carbon-reduction milestones.	High	Pass: The system displays the household's current progress toward its reduction milestone. Fail: Progress is not updated after new emission data is recorded.	To help households monitor whether they are achieving their carbon-reduction goals.
FR-005	Functional	The system shall provide historical comparison of household carbon emissions to support monitoring of reduction progress.	Medium	Pass: Previous and current emission values can be compared. Fail: Historical emission information cannot be accessed or compared.	To help users identify emission trends and evaluate their reduction progress.
NFR-001	Non-Functional – Performance & Security	The annual carbon reduction milestone chart shall render interactive historical comparison graphs in under 200 ms.	High	Pass: Benchmarking tests confirm the target latency and security standards under simulated peak load.	To ensure responsive and secure visualization of historical carbon-reduction data.
NFR-002	Non-Functional – Security	The system shall protect household carbon-footprint and activity information from unauthorized access.	High	Pass: Only authorized users can access household carbon data. Fail: Unauthorized users can access protected information.	To maintain the privacy and security of household sustainability data.

2. UML Use-Case Diagram

Actors: Resident and Sustainability Coach.

Suggested use cases: Log Monthly Utility Consumption, Log Commute Data, Calculate Monthly CO_2e, Track Reduction Milestones, View Historical Emissions, and Review Household Progress.



3. Use-Case Flow Specification

Use Case: Calculate Monthly CO_2e

Primary Actor: Resident

Goal: Calculate the household's total monthly carbon emissions from recorded activity data.

Preconditions

- The Resident has access to the system.
- Monthly electricity and transport data is available.
- Required emission conversion factors are available.

Postconditions

- Monthly CO_2e emissions are calculated.
- The calculated emission value is displayed to the Resident.
- The result is available for tracking reduction milestones and historical comparison.

Main Success Scenario

1. Resident enters monthly electricity consumption in kWh.
2. Resident enters monthly commute distance in km.
3. System validates the entered values.
4. System applies the appropriate conversion factors.
5. System calculates the total monthly CO_2e.
6. System displays the calculated emissions.
7. System updates the household's emission-tracking information.

Alternate Flow – Invalid Data

8. System detects missing, zero, negative, or invalid activity data.
9. System displays an error message.
10. Resident corrects the entered values.
11. System recalculates the monthly CO_2e.